

Operations Manual

How to prepare, run and recover a Boaty session safely: procedures, checklist, contingency cards and the crew card

Document	BOATY-OPS-001
Issue	Issue D (for review)
Date	30 September 2026
Status	For review by the project owner (who is also its main user)
Basis	SRS Issue F (OPS-001 to OPS-012), ConOps/ConUse, FMEA Issue D actions A-05, A-09, A-11, A-17, A-19
Content	14 procedures, 18-item checklist, 12 contingency cards

Document control

Issue	Date	Change	By
A	28 September 2026	First issue.	Claude (drafted)
B	30 September 2026	Simulator findings: K-03 (a dead motor looks like weed), K-05 (the boat stops itself 30 s or 10 m outside the fence), K-11 (a heading alarm in strong wind may be wind drift).	Claude, from simulator slice 2
C	29 September 2026	Mission Control findings: CL-13 (arm at the jetty: home is taken where the boat is armed) and CL-18 (close any other ground station) added.	Claude, from simulator slice 3
D	30 September 2026	SDR decision CR-07: site drawing now puts a ≥ 15 m circle round every nest structure; bird photos only from outside it (OPS-005, SRS Issue G).	Claude, owner decision

Role	Name	Signature / date
Author	Claude, for Project Boaty	
Reviewer / approver	Project owner	

Review guidance: read it as if it's your first lake session. Anything you wouldn't actually do, or that's missing, is a finding. Sections 7 and 8 are meant to be printed and laminated.

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1. About this manual

This is the manual for using Boaty, from the day before a session to putting the battery away. It turns the operations requirements (OPS-001 to OPS-012) and the procedural FMEA actions into things to do, in order. It assumes the boat and software meet their specifications. Where software helps (the checklist, alarms, spoken messages), the procedure says so.

- **Section 5, the pre-launch checklist**, is the same list Mission Control shows before it allows arming (MCN-D13). Edit it here, and both change.
- **References** in brackets (e.g. OPS-004, A-17) point to the requirement or FMEA action each step satisfies. That tells you why a step exists before you're tempted to skip it.
- **Sections 7 and 8** (crew card, quick reference) are one page each, for printing.

2. Roles and golden rules

Role	Responsibilities
Operator (adult)	In charge of safety from packing to unpacking. Holds the magnetic arming key and the PIN. Is the lookout during every mission. Makes every go / no-go call.
Crew (age 4)	Builds and decorates the boat, gives the orders with TALK, presses GO, COME HOME and STOP, spots the boat, finds the ducks. Never goes near the water's edge alone.

1. People first, boat second. If anyone could get hurt, forget the boat. It floats and it can wait. (OPS-007)

2. Nobody goes in the water. Not for the boat, not for anything. Use the pole, the net or the casting line. (OPS-008)

3. Key out before hands on. Pull the magnetic key before anyone touches the boat, and keep it until the boat is on the water. (A-09)

4. The operator is the lookout. Eyes on the boat and the water for the whole mission. If in doubt, STOP. (A-17, OPS-004)

5. Never on shared water without permission. Milton means the Trust's permission and its conditions. (OPS-002)

3. Kit lists

	Group	Item
☐	Boat	Boat (hulls, pods fitted, mast, flag)
☐	Boat	DUPLO crew
☐	Boat	Magnetic arming key on its lanyard (operator's pocket)
☐	Bank station	Mission Control box
☐	Bank station	USB-C power bank (charged)
☐	Bank station	Phone + USB cable (tether and screen)
☐	Bank station	Antenna pole kit (if V-08 showed it's needed)
☐	Battery	Boat battery in a fire-safe bag, charged the day before, carried separately from the boat
☐	Recovery kit	Telescopic pole ≥ 3 m with landing net
☐	Recovery kit	Casting rod with weighted float line
☐	Recovery kit	Throw towel
☐	Spares kit	Spare props
☐	Spares kit	Spare 20 A fuse
☐	Spares kit	Cable ties
☐	Spares kit	Microfibre lens cloth
☐	Spares kit	Towel
☐	Spares kit	Small screwdriver (adult use)
☐	People	Drinks and snacks
☐	People	Hand gel or wipes (algae)
☐	People	Sun hats / waterproofs

4. Procedures

OP-01 Setting up a new site (once per site)

When	Before the first session at any water	Who	Operator
Make sure the fence matches the real water, and that we're allowed and welcome there.			
#	Step	Why	
1	Get written permission from the landowner. At Milton, email Cambridge Sport Lakes Trust and record their conditions in the site notes.	OPS-002	
2	Draw the site on the map at home: inclusion fence 5 m inside the waterline, home points, areas and landmarks.	FEN-002, FEN-003	
3	Add exclusion zones over islands, reed beds and angling swims (≥ 20 m clearance). Put a circle of at least 15 m radius round every nest structure (duck house, nest raft) and tag it wildlife = nest; the site check refuses less. Photos of birds are taken only from outside these circles.	OPS-004, OPS-005	
4	Keep the first fence within 100 m of the launch point. Grow it only after 5 successful missions logged at this site.	OPS-006	
5	Fence survey walk. On the first visit, walk the launch bank carrying the boat's GNSS (or the phone) with logging on. Compare the logged track with the drawn waterline. If they differ by more than 3 m, shift the fence and record the offset in the site file.	A-05	
6	Mark launch points and the wind directions each suits (wind blowing towards that bank).	A-19	
7	Run the site linter. It must pass before the site can be used.	MCN-D53	
Redo the survey walk if the water level has changed a lot or the boat ever reaches the bank inside its fence.			

OP-02 Trial progression

When	Once, from first power-up to first lake mission	Who	Operator
Never find a problem on the lake that we could have found at home.			
#	Step	Why	
1	Simulation: every failsafe scenario passes (SC-01 to SC-13), and the FMEA scenarios run.	OPS-009, SWE-005	
2	Bench rigs L1 and L2: STOP, key, ESC signal loss and brownout tests pass.	OPS-009	
3	Paddling pool: floats level, drives, stops, stuck detection, photo sync, crew controls.	OPS-009	
4	Small private water (if available): first autonomous missions within casting range.	OPS-009	
5	Lake: only when every SIM, BENCH and POOL Must requirement has passed, and OP-03 is done.	OPS-009, OPS-012	

OP-03 Rehearsal at home

When	Before the first lake trial, then each spring	Who	Operator (and crew)
Practise every contingency where mistakes are free.			
#	Step	Why	
1	Start the simulator in rehearsal mode with the real panel.	SIM-D05	
2	Work through every contingency card (section 6) at least once: recognise it, act, recover.	OPS-012	
3	Let the crew practise TALK, GO, COME HOME and STOP on the virtual lake.	OPS-012	

OP-04 The day before

When	Every session	Who	Operator
Arrive with a healthy battery, the right software and the right kit.			

#	Step	Why
1	Battery inspection: no swelling, dents, wrapper damage or smell. Cell voltages balanced within 0.05 V. Pack at room temperature. Any doubt: don't use it.	A-11, PWR-D20
2	Charge on a balance charger, in the fire-safe bag, on a non-flammable surface, attended. Never charge in the boat.	OPS-010, PWR-007
3	Software: use the tagged release only. Run the simulator suite if anything changed.	SWE-006
4	Check the forecast: wind at the lake $\leq$ Beaufort 3, no storms.	OPS-001
5	Pack the kit lists (section 3), including the recovery and spares kits.	OPS-008, OPS-011

OP-05 Transport

When	Every session	Who	Operator
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Arrive with nothing broken or live.

#	Step	Why
1	Battery in its fire-safe bag, separate from the boat.	PWR-007
2	Magnetic key in the operator's pocket, never on the boat.	A-09, MOD-003
3	Boat in its cradle with the mast removed.	MEC-014

OP-06 Setting up on the bank

When	Every session	Who	Operator, crew helping
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Decide whether today is a go, and set up safely.

#	Step	Why
1	Read the notice boards: no swim session, no watersports course, no algae warning. If any, go home and try another day.	OPS-003
2	Look at the water: who else is here? Keep $\geq$ 20 m from anglers' lines and other users.	OPS-004
3	Check the wind. Choose the launch point the site file suggests for today's wind, so a stopped boat drifts to us.	A-19, OPS-001
4	Set up Mission Control a few metres back from the edge. Plug in the phone and start the app.	OPS-007
5	Crew fits the pods and the DUPLO crew. Key stays in the operator's pocket.	A-09, CHD-004
6	Fit the battery, close and latch the box, main switch on.	MEC-012
7	Work through the pre-launch checklist on screen (section 5). Every item must be ticked before arming is possible.	OPS-001, PRE-006

OP-07 Crew briefing

When	Every session, before planning	Who	Operator to crew
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Four buttons, four rules, in words a 4-year-old knows.

#	Step	Why
1	Show the four buttons (see the crew card): blue TALK to tell the boat, green GO to start, yellow house to come home, red STOP.	CHD-005
2	Rule 1: we stay behind the line (a rope or bag marks it).	OPS-007
3	Rule 2: only grown-ups touch the boat when it's on the water.	A-09
4	Rule 3: if you're worried, press the red button. Nobody will be cross.	FS-009
5	Rule 4: we never go in the water, even for the boat.	OPS-008

OP-08 Planning the mission

When Every mission

Who Crew speaks, operator approves

Turn the crew's idea into a safe plan.

#	Step	Why
1	Crew holds TALK and says what the boat should do.	NLI-002
2	Listen to the summary and look at the map: duration, distance, photo count, furthest point from home.	MCN-D56, NLI-005
3	Check the route stays well clear of other people and wildlife. If anyone is within 50 m of the route, change the plan or wait.	A-17, OPS-005
4	Approve with the PIN (never read the PIN aloud). No internet: choose a template with the crew.	VAL-008, MIS-005

OP-09 Launch and arming

When Every mission

Who Operator

Get the boat on the water with nobody's hands near a prop.

#	Step	Why
1	Carry the boat by the handle, put it on the water at the chosen launch point, and step back.	A-09
2	Only now insert the magnetic key into its dock.	A-09, MOD-003
3	Arm at Mission Control. Check home is shown in the right place.	MOD-003, FM-46
4	Crew holds GO for a second: 'Off we go!'	MC-004

OP-10 During the mission: the lookout

When Whole mission

Who Operator

Be the eyes the boat doesn't have.

#	Step	Why
1	Watch the boat and the water around it all the time. Hand the phone to nobody.	A-17
2	Anyone (swimmer, paddler, dog, angler's line) within 20 m of the boat or 50 m of its route: press COME HOME, or STOP if closer.	OPS-004, A-17
3	Birds with young, or the boat heading for reeds or nests: COME HOME.	OPS-005
4	Keep the crew behind the line. Being with the child comes before watching the boat.	OPS-007
5	Anything strange (circling, wrong direction, alarms): STOP first, think second (section 6).	FM-17, FM-05

OP-11 Recovering the boat

When End of every mission

Who Operator

Get the boat out with the motors dead.

#	Step	Why
1	Boat waits by the bank at home (HOLD). Disarm at Mission Control.	MOD-005
2	Pull the magnetic key and pocket it, then lift the boat by the handle or hoop. Never by a pod.	A-09
3	Look at the guards and props: cracked guard = no more missions today.	A-09, MEC-010

OP-12 After the session

When Every session

Who Operator, crew for photos

Leave the boat, battery and records ready for next time.

#	Step	Why
1	Let photos sync and make the captain's log together.	DET-002
2	Rinse pods and hulls with clean water and dry the motors. Everyone washes hands.	OPS-003
3	Main switch off only after Mission Control says the mission computer has shut down.	IF-08
4	Battery out, home in the fire-safe bag. Within a day, storage-charge it (≈ 3.7 V/cell).	OPS-010
5	Download logs. Note anything odd as an incident (OP-14).	LOG-003

OP-13 Maintenance

When	See table	Who	Operator
Catch wear before it catches us.			
Interval	Tasks	Refs	
Every session	Rinse and dry pods and motors; inspect guards; check box seal; battery inspection (OP-04)	A-09, A-11	
Monthly (in season)	Replace worn props; check thumb-screws and latches; clean camera window; check the SD card health report	MEC-007	
Each season	Box dunk test; re-measure pack capacity (PWR-D21); change the PIN; re-run the full simulator suite; review the FMEA	PWR-D21, SAF-006, MCN-D61	
After any change	New firmware, parameters or code: tagged release, full simulator suite, rig retest of anything touched	SWE-006, SAF-007	

OP-14 Incidents and learning

When	Whenever something unexpected happens	Who	Operator
Every surprise makes Boaty safer.			
#	Step	Why	
1	Write down what happened, when, and what the logs show.	LOG-003	
2	Check it against the FMEA. New failure mode or worse than rated? Update the FMEA and add a test.	SAF-006	
3	Anything involving people or wildlife: pause lake sessions until it's understood.	OPS-004, OPS-005	

5. Pre-launch checklist

Every item must be ticked before Mission Control allows arming (PRE-006). 'Automatic' items are ticked by the system and can't be overridden. Record of who signed off and when is kept in the session log (MC-010).

ID	Group	Check	How	Why
☐ CL-01	Site and people	Permission held and today's conditions met	Operator	OPS-002
☐ CL-02	Site and people	No swim session, watersports course or algae warning	Operator	OPS-003
☐ CL-03	Site and people	Nobody within 50 m of the planned area; anglers' swims excluded	Operator	OPS-004, A-17
☐ CL-04	Site and people	Wildlife exclusions reviewed on the map	Operator	OPS-005
☐ CL-05	Weather	Wind Beaufort 3 or less (small waves, few white horses)	Operator	OPS-001
☐ CL-06	Weather	Wind blowing towards our bank; launch point chosen	Operator (suggested by the app)	A-19, OPS-001
☐ CL-07	Boat	Battery inspected and $\geq 80\%$ charged	Operator + automatic	A-11, PRE-003
☐ CL-08	Boat	Box sealed and latched; glands tight	Operator	OPS-001, MEC-012
☐ CL-09	Boat	Guards intact; props free; pods latched	Operator	A-09, MEC-010
☐ CL-10	Boat	Flag, hoop and beacon fitted and working	Operator	OPS-001, REC-003
☐ CL-11	Boat	Key-out rail test passed	Automatic	MCN-D54
☐ CL-12	Bank	Button test passed (TALK, GO, COME HOME, STOP)	Automatic	MCN-D55
☐ CL-13	System	Fence and exclusions reviewed; boat at the jetty when armed (the helm takes home from where it is armed; Mission Control refuses a plan more than 10 m from the site's home)	Operator	OPS-001, PRE-002, VAL-004
☐ CL-14	System	Parameters match baseline; ≥ 8 satellites; link good	Automatic	SAF-007, PRE-001
☐ CL-15	Recovery	Recovery kit on the bank (pole and net, casting rod and line)	Operator	OPS-008, OPS-001
☐ CL-16	Recovery	Spares kit on the bank	Operator	OPS-011
☐ CL-17	People	Adult operator present and acting as lookout; crew briefed	Operator	OPS-007, A-17
☐ CL-18	System	Anything else talking to the boat closed (QGroundControl, a second tablet): Mission Control refuses to arm while it hears one	Operator	MCN-D57

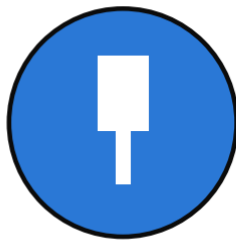
6. Contingency cards

What to do when something goes wrong. The first move is almost always the same: **people first, then STOP or COME HOME, then think.**

K-01 Link lost	
You'll notice	'Link lost' on screen and spoken; last position frozen on the map
Operator	Stay put or move to a better spot. In a mission, the boat carries on, then comes home by itself after 60 s. Driving manually, it stops, then comes home after 10 s.
Crew	Watch for the flag
Refs	FS-002, FS-003
K-02 Low battery	
You'll notice	'Coming home: battery low'
Operator	Let it come home. End the session.
Crew	Wave it in
Refs	FS-001
K-03 Stuck in weed	
You'll notice	'I'm stuck, trying to wiggle free'
Operator	Wait for up to three wiggles. Still stuck: COME HOME to retry, else recover by line (K-07). Stuck in clear water with no weed in sight? Suspect a dead motor: the boat can't tell the two apart. Recover and inspect the pods.
Crew	Spot the boat
Refs	FS-005, FS-006, FM-50
K-04 Boat stopped: position lost	
You'll notice	'I've lost my way, stopping'
Operator	Wait up to a minute. It goes home once GPS is back. If not, recover (K-07).
Crew	-
Refs	FS-004
K-05 At or beyond the fence	
You'll notice	Fence alarm; boat heading home
Operator	Let it return. If wind keeps it outside, it stops itself after 30 s or 10 m out ('outside fence' alarm): recover it (K-07). If it's pushed against the bank, STOP and recover.
Crew	-
Refs	FEN-005, FEN-006, MCP-D30
K-06 Person or animal near the boat	
You'll notice	You see them first: the boat can't
Operator	COME HOME, or STOP if close. Wait until clear.
Crew	Press STOP if asked
Refs	OPS-004, A-17

K-07 Boat dead in the water	
You'll notice	Stopped; no response; beacon may be dark
Operator	It floats. Wait for the wind to bring it to our bank. Pole and net if within reach. Otherwise cast the weighted line beyond the boat and reel it in so the line catches the mast and hoop. Never wade.
Crew	Stay behind the line
Refs	OPS-008, REC-005
K-08 No internet	
You'll notice	'Templates only'
Operator	Choose a template with the crew.
Crew	Pick an adventure
Refs	MIS-005
K-09 Mission Control dead	
You'll notice	Screen and buttons dead
Operator	The boat comes home by itself after 60 s. Restart the box. Recover as normal.
Crew	-
Refs	FS-003
K-10 Water or heat in the box	
You'll notice	'Water in the box' or 'too hot' alarm; boat returning
Operator	Let it return, key out, battery out, dry or cool, and investigate before the next mission.
Crew	-
Refs	FS-010, A-11
K-11 Boat behaving strangely	
You'll notice	Circling, going the wrong way, or a navigation alarm
Operator	STOP. Look before resuming: pods, weed, compass. A 'heading check' alarm in strong wind may be the wind blowing the boat backwards, not the compass: check the wind first. Resume only with the PIN if you understand why.
Crew	Press STOP if asked
Refs	FM-17, FM-05, FM-49, MCN-D60
K-12 Battery hot, swollen, smoking or smelling	
You'll notice	Heat, hissing, smoke
Operator	Keep everyone away, especially the crew. Don't touch it. If it's on land and burning, let it burn out on a non-flammable surface or cool it with lots of water from a distance. Call 999 if fire spreads.
Crew	Go with a grown-up, away from it
Refs	FM-23, OPS-010

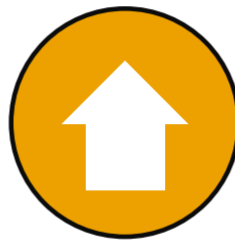
7. Crew card (print me)



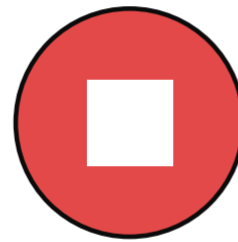
TALK



GO



COME HOME



STOP

Stay behind the line · Only grown-ups touch the boat · Never go in the water

Button	Looks like	What it does
TALK	Blue, microphone	Hold it and tell Boaty what to do.
GO	Green, arrow	Hold it to start. Only when the grown-up says.
COME HOME	Yellow, house	Boaty comes back to us.
STOP	Red, square	Boaty stops. Press it any time you're worried.

Read it together before every session (OP-07). The crew can't break anything by pressing a button: GO only works when the grown-up has made it safe.

8. Quick reference (print me)

Golden rules

1. People first, boat second.
2. Nobody goes in the water.
3. Key out before hands on.
4. The operator is the lookout.
5. Never on shared water without permission.

A session in eight lines

OP-04 Day before: inspect and charge battery, pack kits

OP-06 Bank: notices, other users, wind → launch point

Checklist (section 5) → OP-07 crew briefing

OP-08 TALK → check plan → approve with PIN

OP-09 Boat on water → step back → KEY IN → arm → GO

OP-10 Lookout the whole time → STOP if in doubt

OP-11 Home → disarm → KEY OUT → lift by handle

OP-12 Sync photos, rinse, wash hands, storage-charge

If...	Do
Link lost	Stay put or move to a better spot.
Low battery	Let it come home.
Stuck in weed	Wait for up to three wiggles.
Boat stopped: position lost	Wait up to a minute.
At or beyond the fence	Let it return.
Person or animal near the boat	COME HOME, or STOP if close.
Boat dead in the water	It floats.
No internet	Choose a template with the crew..
Mission Control dead	The boat comes home by itself after 60 s.
Water or heat in the box	Let it return, key out, battery out, dry or cool, and investigate before the next mission..
Boat behaving strangely	STOP.
Battery hot, swollen, smoking or smelling	Keep everyone away, especially the crew.

Appendix A. Traceability

Requirement	Text (abridged)	Met in
OPS-001	The pre-launch checklist shall cover at least: site permission and conditions; no swim session, course or al...	OP-04, OP-06, CL-05, CL-06, CL-08, CL-10, CL-13, CL-15
OPS-002	Written permission from Cambridge Sport Lakes Trust shall be obtained before any operation at Milton Country...	OP-01, CL-01
OPS-003	There shall be no operation during swim sessions or watersports courses, or while a blue-green algae warning...	OP-06, OP-12, CL-02
OPS-004	Exclusion zones shall keep the boat at least 20 m from occupied angling swims and other water users. If anyo...	OP-01, OP-06, OP-10, OP-14, CL-03, K-06
OPS-005	Islands, reed beds and known nesting areas shall be exclusion zones, and every nest structure shall have an ...	OP-01, OP-08, OP-10, OP-14, CL-04
OPS-006	At Milton, the Mk1 inclusion fence shall stay within 100 m of the launch point until at least 5 successful l...	OP-01
OPS-007	An adult operator shall be present and in control throughout. The child shall never be unsupervised near the...	OP-06, OP-07, OP-10, CL-17
OPS-008	The recovery kit shall include a telescopic pole of at least 3 m with a landing net, and a casting rod with ...	OP-04, OP-07, CL-15, K-07
OPS-009	Trials shall progress: simulation, then bench, then paddling pool or small water, then lake. Each failsafe s...	OP-02
OPS-010	Batteries shall be charged under supervision in a fire-safe bag, and stored at storage voltage.	OP-04, OP-12, K-12
OPS-011	A spares kit shall be taken to every session: spare propellers, a spare fuse, cable ties, a microfibre lens ...	OP-04, CL-16
OPS-012	Before the first lake trial, the operator shall rehearse every contingency scenario in section 3.5 in simula...	OP-02, OP-03
FMEA action	Action	Met in
A-05	Fence survey procedure: walk the launch bank with a GNSS reference to check imagery offset before the first mission at a site	OP-01
A-09	Checklist and procedure: key out before anyone touches the boat; lift only by the handle or hoop; inspect guards	OP-05, OP-06, OP-07, OP-09, OP-11, CL-09
A-11	Box temperature sensor with alarm and RTL above 60 °C; battery inspection before each session	OP-04, CL-07, K-10
A-17	Operator is the dedicated lookout; brief 'STOP if in doubt'; no missions while people are within 50 m of the planned route	OP-08, OP-10, CL-03, CL-17, K-06
A-19	Site file lists the preferred launch point per wind direction; the checklist asks 'is the wind blowing towards us?' and suggests the launch point	OP-01, OP-06, CL-06